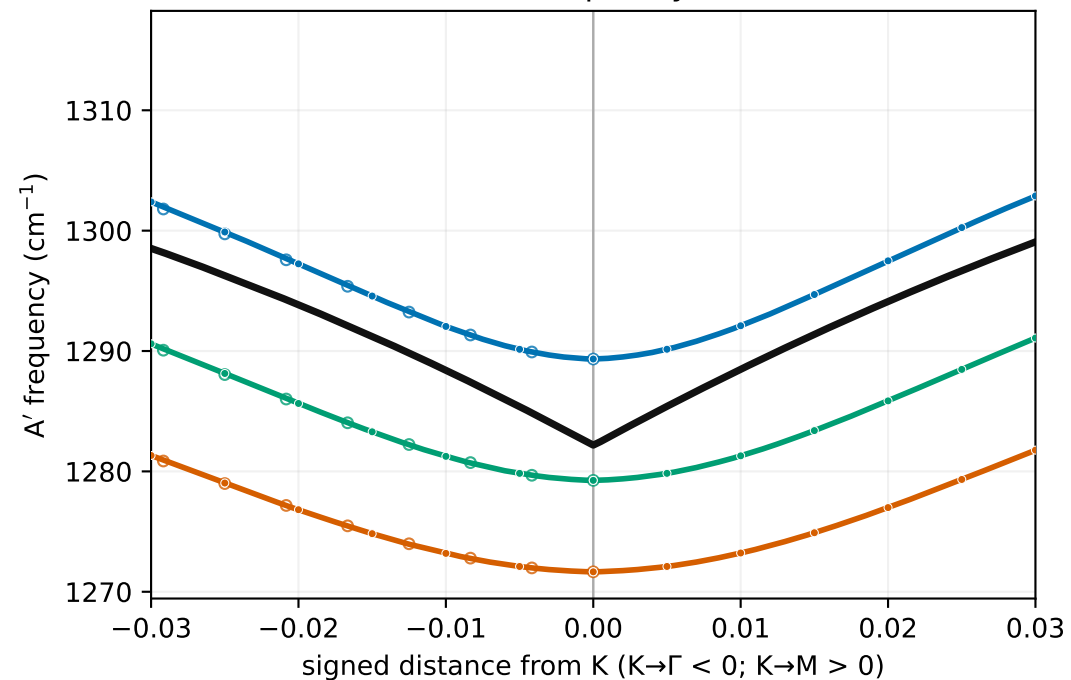
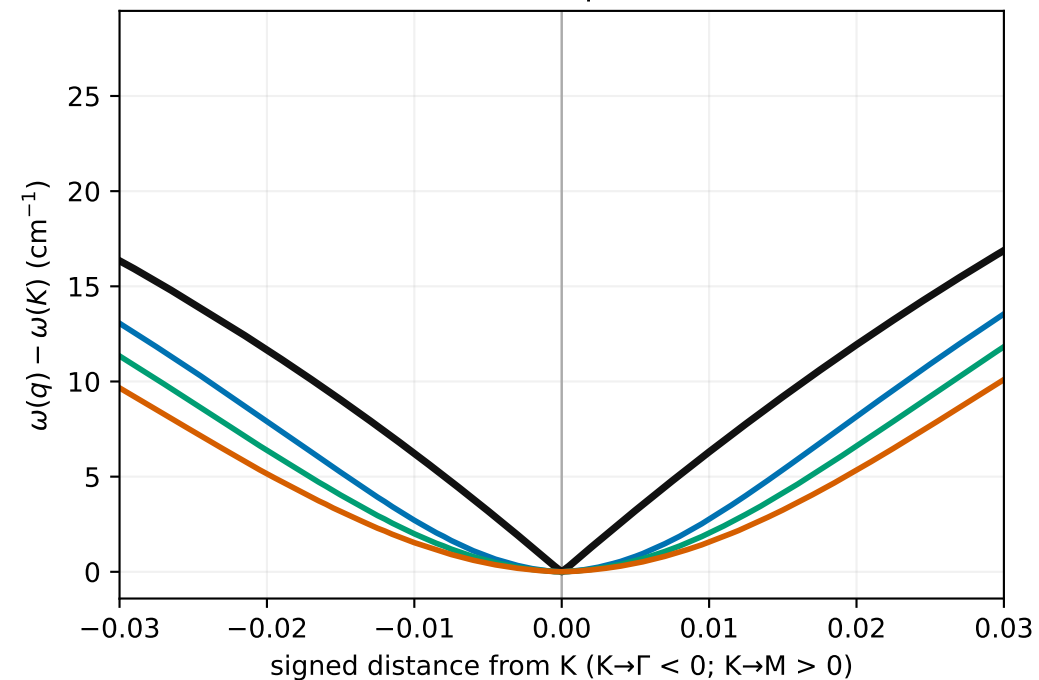


Graphene K-neighbourhood A': MLIP short range + full EPC long range

Absolute frequency: Γ -K-M



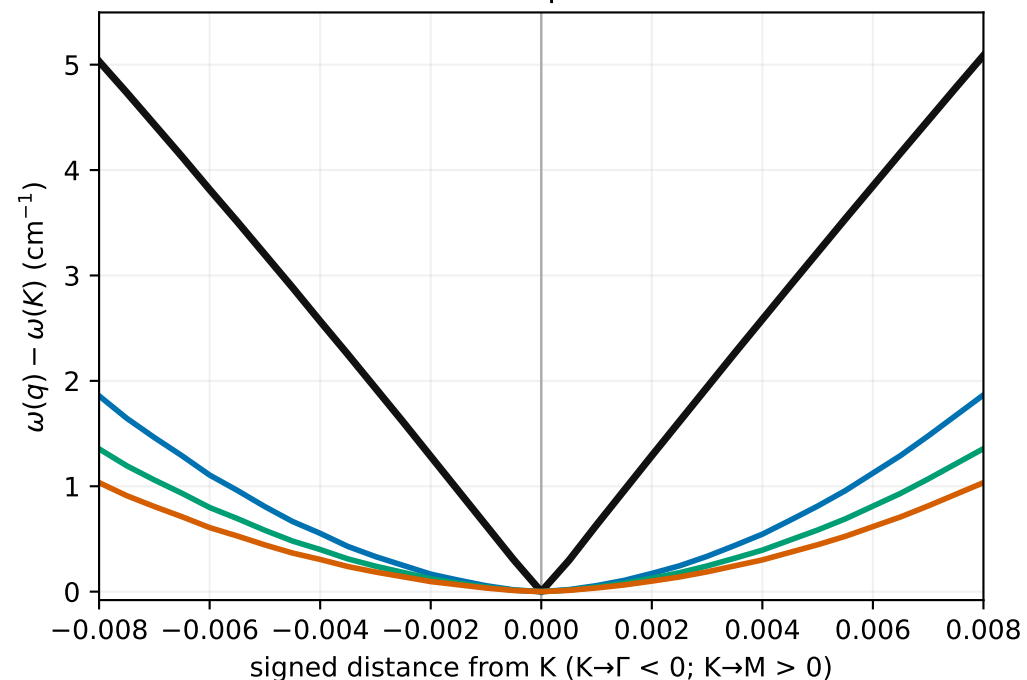
K-referenced cusp: full window



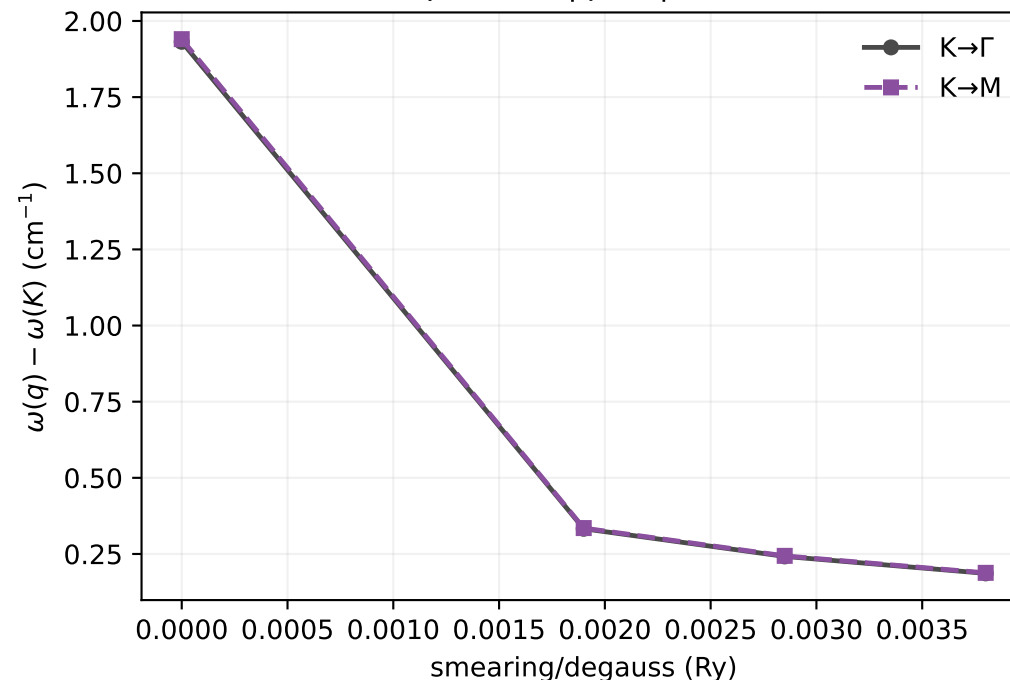
smearing/degauss (Ry)

- 0 Ry
- 0.0019001 Ry; lattice 300 K
- 0.0028501 Ry; lattice 450 K
- 0.0038002 Ry; lattice 600 K

K-referenced cusp: inner window



Cusp rise at $|q - K| = 0.003$



Finite curves: L0 short-range backgrounds + direct full-EPC q-space response.
 Filled dots: calculated q samples.
 Open dots: existing nk=720 $K \rightarrow \Gamma$ replay.
 $K \rightarrow M$ is calculated directly, not mirrored.

Absolute finite-smearing shifts also include the paired 300/450/600 K lattice backgrounds; use the K-referenced panels for cusp shape.